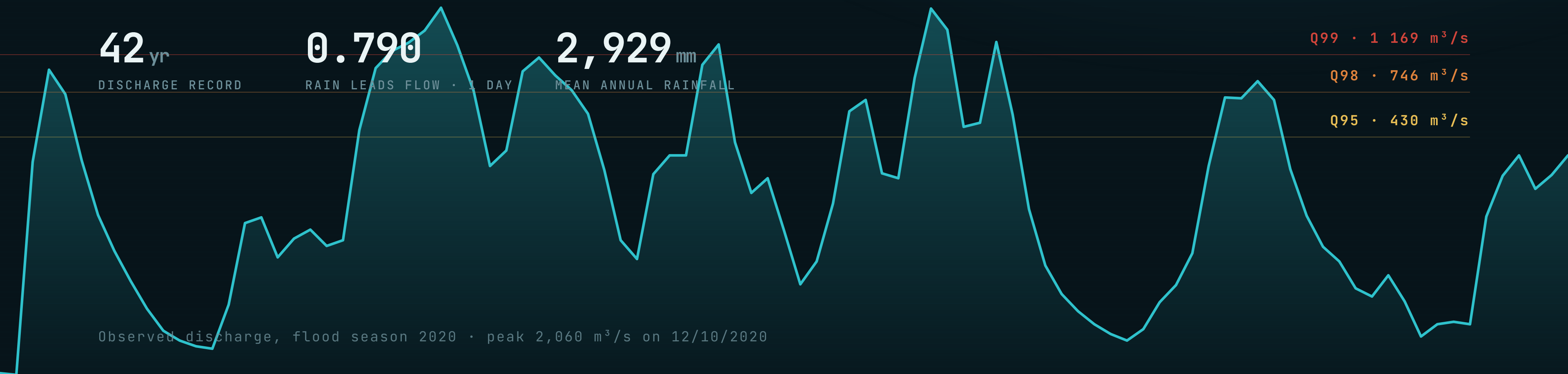


Three days of warning

Flood risk for the Hương River, Huế — forecast from open data and reported in Vietnam's own alert stages.



— SUBJECT & CONTEXT

A basin that floods in a single day

- Central Vietnam is the country's most flood-affected region.
- Short, steep catchment — rain becomes a flood peak within **one day**.
- Commune-level warning remains sparse relative to the damage.

OFFICIAL ALERT STAGES

Kim Long station · water level above datum

BĐ I		1.00 m
BĐ II		2.00 m
BĐ III		3.50 m

Decision 05/2020/QĐ-TTg — cross-checked against five bulletins of the Huế Hydro-Meteorological Station.

Not the volume — the variety and the veracity

Volume

42 years of daily discharge, 11 years of hourly rainfall, 68 grid points. About **1,500 Parquet files** — moderate, not extreme.

Variety

Six source types on one daily timeline: simulated flow, reanalysis rain, archived forecasts, river geometry, a legal document, news bulletins.

Velocity

Forecasts are re-issued several times a day. An automated job pulls a fresh 7-day forecast **every morning**.

Veracity

Discharge is **model output, not measurement**. Reservoir releases are unmodelled. Stated, not hidden.

Two targets, one awkward bridge

01 How much water?

Daily discharge in m^3/s at 1, 2 and 3 days. Continuous target – RMSE, MAE, NSE, KGE.

02 Which alert stage?

Exceedance of BD I / II / III. Positive days are **1–3 %**, so accuracy is banned – POD, FAR, CSI instead.

03 The bridge between them

Stages are **metres**; our data is m^3/s . No measured rating curve was available, so the mapping is built from documented flood peaks.

— ANALYTICS APPROACH

Primarily predictive

SUPPORTING

Descriptive

Seasonality, rainfall–discharge lag, historical events.

PARTIAL

Diagnostic

SHAP attributes each warning to rain, wetness or season.

CORE

Predictive

Discharge and alert stage 1–3 days ahead. Both research questions.

OUT OF SCOPE

Prescriptive

Evacuation decisions need authority we do not hold.

We do not replace the global forecast. We **correct it locally** and **translate it** into Vietnam’s alert stages at sub-basin scale.

— DATA REQUIREMENTS

Six inputs, one daily table

Dataset	Role	Span	Status
River discharge — GloFAS v4	Target and lag features	1984–2026, daily	Collected
Rain & meteorology — ERA5	Primary predictors	2010–2026, daily + hourly	Collected
Archived rainfall forecasts	Operational input scenario	2022–2026, daily	Collected
River centre-lines — OpenStreetMap	Grid cell to river	Current extract	Collected
Legal alert thresholds	Defines the target	05/2020/QĐ-TTg	Verified
Documented flood peaks	Calibrates metres to m³/s	≥ 15 events	In progress

The last row is the **critical path**. Without it the alert-stage target cannot be defined at all.

— COLLECTION METHOD

How the data gets here

1

Select

Scan 361 cells; keep the one whose inferred catchment matches the real basin.

→

2

Crawl

Independent tasks, one file each. Retry, cache, checkpoint, lock.

→

3

Clean

Normalise to Vietnam time, hourly→daily, validate and fail loudly.

→

4

Assemble

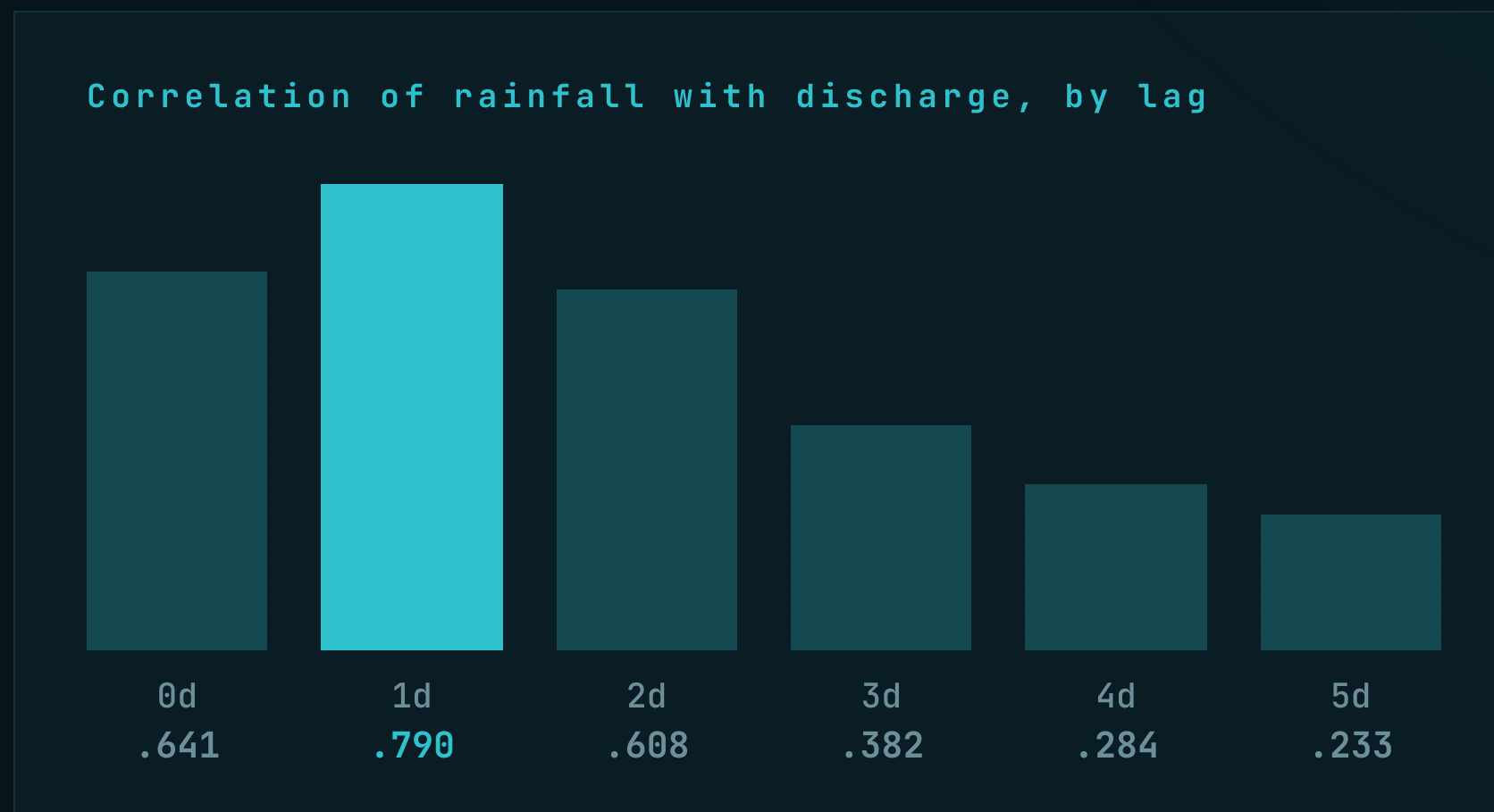
Average rain over three sub-basins, build lags, block leakage.

WHAT WE GOT WRONG

Persistent rate-limit errors were blamed on the provider — and a scope cut was proposed. The real cause was **three of our own crawlers running in parallel**. With one process the whole collection took **35 minutes, zero errors**. Lesson: count your own processes first.

— FEASIBILITY — ALREADY DEMONSTRATED

The signal is measurable



- **6,087 × 71** daily table, 0 % missing discharge.
- Rainfall **2,929 mm/yr** matches Huế's published climatology — an independent check on our own data.
- Single-day rain beats 3-, 5- and 7-day accumulations: the basin **responds fast**.
- Baselines measured, so the bar is real: persistence reaches **NSE 0.542** at one day, then collapses.

— IN ONE SENTENCE

We correct and localise an open global flood forecast into Vietnam's own alert stages.

NEXT THREE STEPS

1. Complete the flood-event record that unlocks the alert-stage target.
2. Train the model and beat the measured baselines.
3. Test whether reservoir operation explains the anomalies we found.

STATED PLAINLY IN THE REPORT

- Discharge data is **simulated, not measured**.
- The 5 km grid cannot resolve the neighbouring Bồ River — so the study covers **one station**.
- Academic work, **not an official warning service**.

Thank you — questions welcome.